CampusSpace

Class & Lecture Hall Management System — PRD

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PROBLEM ALIGNMENT

Indian universities have no reliable way to manage 200+ teaching spaces — scheduling is done entirely on Excel/Google Sheets with no conflict detection, no real-time visibility, and no audit trail. Double-bookings surface only when faculty physically arrive at a room, forcing last-minute fixes that cascade through the day. Timetable admins rebuild the full semester schedule manually every term — a process that takes **2–3 weeks but should take 2 days**. At 15,000 students, 500 faculty, and 200 rooms across 6 days a week, the coordination cost of doing this offline is structurally unsustainable.

Assumes: Mon–Sat 8 AM–6 PM schedule • 2–5 person admin team • SIS read-only access • batch sizes 25–130 • structured CSV available day-one • WhatsApp/SMS preferred for alerts • limited building accessibility (RPwD Act 2016 relevant).

HIGH LEVEL APPROACH

A centralised, web-based scheduling layer that sits on top of the existing SIS — giving timetable admins a single conflict-aware grid to allocate rooms, and giving faculty and students read-only visibility into their schedules. The core bet: if conflict detection is built into the act of booking itself, double-bookings become **structurally impossible — not just less likely.**

NARRATIVE

- **Priya — Timetable Admin:** Sole coordinator for a 15,000-student university.

Opens a 200-row Google Sheet each semester, fills allocations manually → By day three, two departments edit simultaneously → LH-201 is double-booked Tuesday at 10 AM → She finds out on Tuesday at 10:05 AM, when two faculty members call her at the same time

- **Dr. Mehta — Faculty:** Teaches Digital Electronics to 90 students.

Arrives at SR-101 to find another class in session → Checks WhatsApp — no update → Walks to admin office, waits 10 minutes, redirected to a room that fits 40 → He runs the class anyway → 50 students stand

- **Students (15,000 across the university):** Want to know where their next class is, especially when rooms change.

Currently rely on faculty WhatsApp groups, class reps, and noticeboards → MVP solves this via the published shareable link (F9) → dedicated student view is Phase 3

GOALS

G1. Reduce timetable build time from 2–3 weeks to under 2 working days per

semester. **G2.** Zero room/faculty double-bookings at the point of timetable publication.

G3. All capacity overages surfaced before publish — admin signs off on any overrides.

G4. Faculty can check their weekly schedule without calling admin.

G5. Admin publishes the semester schedule with confidence — every conflict caught before it became a problem.

Non-Goals: • SIS replacement • exam scheduling • attendance tracking • native mobile app.

KEY FEATURES

With limited dev bandwidth, MVP is defined by one question: what is the minimum set of features that makes double-bookings structurally impossible and timetable build time under 2 days?

P0 = must-ship. P1 = ships if bandwidth allows.

#	Feature Pri Goal
F1	Room inventory — capacity, type, facilities, floor, accessibility P0 G2, G3
F2	Timetable grid — room × day × time slot allocation by admin P0 G1, G2
F3	Real-time conflict detection — room double-book + faculty double-book P0 G2
F4	Capacity warning — alert when enrollment exceeds room capacity P0 G3
F5	SIS read integration — import courses, enrollment counts, faculty P0 G1
F6	CSV import — migrate existing Google Sheet timetable on day one P0 G1
F7	Faculty read portal — own weekly schedule, no login friction P1 G4
F8	Room search + filter — type, capacity, building, availability P1 G1
F9	Timetable publish + shareable link P1 G4, G5

***Conflict + capacity rules**: room/faculty double-book blocks on confirmation; capacity overage warns (escalated at 20%+); admin override logged. Parallel sections (same course, two rooms, same slot) are valid — not a conflict. Single-admin in MVP — concurrent edits last-write-wins; multi-admin locking is Phase 2. Full rules in supporting doc.*

KEY TRADEOFFS

- **Warn vs. block on capacity overage:** admin keeps autonomy (wins), students in oversubscribed rooms may stand (lose). Decision: warn + log override; RPwD warning escalates at 20%+.
- **Single-admin MVP vs. multi-admin locking:** solo admins ship faster (wins), 2–5 person admin teams must coordinate offline (lose). Decision: single-admin in MVP, locking in Phase 2.
- **Admin-only edit vs. department coordinator role:** central control prevents conflicts (wins), departments wait on central admin (lose). Decision: admin-only in MVP, HoD read view in Phase 2.

FUTURE CONSIDERATIONS

- **Phase 2 — Operational Layer:** Room change requests · maintenance blocking (date-range) · WhatsApp/SMS alerts · department coordinator role · HoD read view.
- **Phase 3 — Intelligence Layer:** Auto room suggestion · utilisation analytics · student-facing schedule view · exam scheduling module.

KEY FLOWS

Prototype (all flows interactive): https://campusspace-psi.vercel.app/ — Admin, Faculty, Student views via persona switcher.

- **Flow 1 — Admin schedules a room:** Click empty slot → modal pre-fills room/day/time → SIS dropdowns for course + faculty → real-time conflict + capacity check → confirm.
- **Flow 2 — Admin migrates timetable on day one:** Sync SIS (preview + confirm courses) → Import Excel (.xlsx) → system flags conflicting rows → confirm valid rows → timetable live.
- **Flow 3 — Faculty checks weekly schedule:** Select name from dropdown (no login) → 6-column Mon–Sat grid → 'Free' shown for days with no sessions.

LAUNCH PLAN

Phase	Exit Criteria
Pilot — Internal Synthetic data; all P0 flows validated end-to-end	Zero P0 bugs on rolling 5-day basis. SIS import and CSV import complete without errors.
Beta — 1 University Real admin builds full semester schedule	Timetable completed in under 2 working days. Zero room/faculty double-bookings at publish. At least 1 admin rates the product 4+/5 and confirms timetable build time dropped by 50% or more.
Launch — Multi-University 3–5 universities across different SIS providers	Each university onboards in under 1 day. Timetable build time under 2 days in all cases.

KEY RISKS

- **SIS fragmentation:** SAMARTH covers central universities and NITs; hundreds of private institutions run India-built ERPs (Academia, Camu, Infinity) with no standard API. Mitigate: prioritise SAMARTH, build generic CSV fallback.
- **Semester timing lock-in:** Product can only be meaningfully tested at semester start — a delayed beta means a 4–6 month wait. Mitigate: begin outreach 8 weeks before semester start.
- **Admin adoption:** 10-year Excel users resist switching. Mitigate: side-by-side CSV import for frictionless day-one migration.
- **Concurrent admin edits:** MVP is single-admin by design; last write wins. Mitigate: document constraint clearly at onboarding.
- **Peak load at semester start:** 2-day build window means all 200 rooms × 500 faculty hit the system simultaneously. Mitigate: rate-limit SIS sync calls, queue conflict checks at the booking modal.

OPEN QUESTIONS

- **SIS integration method:** API access or CSV export only? Determines F5 build complexity. ● **Holiday calendar ownership:** university admin, registrar, or academic calendar system? Needs a single source of truth.
- **Accessibility data in SIS:** does the SIS carry per-student or per-course accessibility flags? Determines whether RPwD warning can be made smarter in Phase 2.

Support Document : **Full Key Logic, flow walkthroughs, and open questions**